

EXPT NO:6	INTEGRATE DATA FROM VARIOUS SOURCES FOR COMPREHENSIVE ANALYSIS USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is data integration in the context of Business Intelligence (BI)?

Data integration is the process of combining data from multiple sources into a single, unified dataset for analysis. In Business Intelligence (BI), it helps organizations obtain a complete view of business performance by bringing together data from different systems such as databases, CSV files, Excel sheets, cloud services, and ERP systems. This enables better reporting, decision-making, and insights.

2. Explain the difference between a join and a data blend in Tableau.

Join:

- Combines tables at the data source level.
- Creates a single table before analysis.
- Requires a common field between datasets.
- Suitable when data comes from the same database or compatible sources.

Data Blend (Relationship):

- Combines data during visualization.
- Keeps datasets separate and links them using common fields.
- Useful when data comes from different sources.
- More flexible for independent datasets.

3. What are relationships and cardinality (e.g., one-to-many, many-to-many) in Power BI's data model?

A **relationship** connects two tables using a common field so they can work together during analysis.

Cardinality describes how records relate between tables:

- **One-to-One (1:1):** One record matches one record.
- **One-to-Many (1:*):** One record in one table matches many records in another table.
- **Many-to-One (*:1):** Many records connect to one record (commonly used for lookup tables).
- **Many-to-Many (:):** Multiple records in both tables are related.

4. Why is data quality (e.g., consistent naming conventions, data types) crucial when integrating data from different sources?

High-quality data ensures accurate integration and reliable analysis. Consistent naming conventions, correct data types, and clean data prevent errors such as failed joins, duplicate records, incorrect calculations, and misleading visualizations. Good data quality improves the accuracy and credibility of business insights.

5. Name two common challenges encountered during data integration.

- Inconsistent data formats or naming conventions (e.g., "N. America" vs. "North America").
- Missing or duplicate data, leading to inaccurate analysis.

IN-LAB EXERCISE:

OBJECTIVE:

To connect to, combine, and model data from multiple disparate sources in Tableau and Power BI to create a unified, comprehensive dashboard for holistic analysis.

PROCEDURE:

1. Data Collection:

- Import data from at least two different sources (e.g., a CSV file and an Excel file).
- Dataset 1: Sales Data (CSV file) with fields like Order ID, Order Date, Region, Sales Amount.
- Dataset 2: Regional Sales Targets (Excel file) with fields like Region, Sales Target, Target Year.

2. Data Preparation and Transformation:

- Clean both datasets: Check for inconsistencies in region names (e.g., "N. America" vs. "North America").
- Ensure data types are correct (e.g., Sales Amount is numeric, Order Date is a date).
- Handle any missing values as appropriate.

3. Data Integration and Modeling:

- Tableau: Use the Data Source pane to establish a relationship or join between the Sales Data and Regional Targets tables using the common Region field.
- Power BI: Use the Model view to create a relationship between the two tables, connecting them on the Region column. Define the cardinality (e.g., many-to-one).

4. Create Calculated Fields for Comparison:

- Develop new measures that leverage both datasets.

- **Examples: Target Achievement % = (SUM(Sales) / SUM(Sales Target)) * 100**
- **Sales vs. Target Variance = SUM(Sales) - SUM(Sales Target)**

5. Design a Comprehensive Dashboard:

- **Create visuals that combine data from both sources to provide a single view of performance.**
- **Add KPI cards for Total Sales and Overall Target Achievement %.**
- **Use a gauge chart or bullet chart to show sales performance against targets for each region.**
- **Create a combined bar chart showing Actual Sales vs. Target Sales by Region.**
- **Use a map to visualize the Sales vs. Target Variance, highlighting underperforming or overperforming regions.**

6. Validate Data Integrity:

- **Verify that the joined/related data is accurate by cross-checking values.**
- **Ensure filters on the dashboard correctly update visuals that use data from both sources.**

7. Publishing and Sharing:

- **Tableau: Publish the workbook to Tableau Public or Tableau Server.**
- **Power BI: Publish the report to Power BI Service and share the link.**

Experiment Datasets:

Dataset 1: Retail Sales Data (CSV)

Fields Used:

- **Order ID – Unique identifier for each order**
- **Order Date – Date of the sales transaction**
- **Region – Sales region**
- **Sales – Sales revenue from the order**

Dataset 2: Regional Sales Targets (Excel)

Fields Used:

- **Region – Sales region (used as the key for joining/relationship)**
- **Sales Target – The sales goal for the region**
- **Target Year – The year for which the target is set**

TABLEAU STEPS :

Tableau - Book1

File Data Server Window Help

Connections

- Sample - Superstore.csv

Sheets

- Sample - Superstore

Sample - Superstore (Sample - Superstore.csv)

Connection: ☒ Live ☐ Extract Filters: 0 | Add

Sample - Superstore

21 fields 3726 rows

100 rows

#	Sample - Superstore	Sample - Superstore	Sample - Superstore	Sample - Superstore	Sample - Superstore	Sample - Superstore	Sample - Superstore
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	
6269	CA-2015-115392	10-01-2015	10-04-2015	Second Class	RM-19675	Robert Marley	
6270	CA-2017-102379	12-02-2017	12-06-2017	Standard Class	BB-11545	Brenda Bowman	
6271	CA-2017-102379	12-02-2017	12-06-2017	Standard Class	BB-11545	Brenda Bowman	
6272	CA-2017-102379	12-02-2017	12-06-2017	Standard Class	BB-11545	Brenda Bowman	
6273	CA-2017-149706	12-11-2017	12-12-2017	First Class	AS-10285	Alejandro Savel	
6274	US-2016-139087	7/17/2016	7/22/2016	Second Class	DK-13375	Dennis Kane	
6275	US-2016-139087	7/17/2016	7/22/2016	Second Class	DK-13375	Dennis Kane	
6276	US-2016-139087	7/17/2016	7/22/2016	Second Class	DK-13375	Dennis Kane	
6277	US-2016-139087	7/17/2016	7/22/2016	Second Class	DK-13375	Dennis Kane	
6278	US-2016-139087	7/17/2016	7/22/2016	Second Class	DK-13375	Dennis Kane	
6279	US-2016-139087	7/17/2016	7/22/2016	Second Class	DK-13375	Dennis Kane	
6280	US-2016-139087	7/17/2016	7/22/2016	Second Class	DK-13375	Dennis Kane	
6281	CA-2017-152226	05-12-2017	5/16/2017	Standard Class	JH-15910	Jonathan Howel	
6282	CA-2015-116484	11/20/2015	11/26/2015	Standard Class	JK-15205	Jamie Kunitz	

Tableau - Book1

File Data Server Window Help

Connections

- Sample - Superstore.csv
- Regional_Sales_Targets_Final

Sheets

- Regional Sales Targets

Sample - Superstore+ (Multiple Connections)

Connection: ☒ Live ☐ Extract Filters: 0 | Add

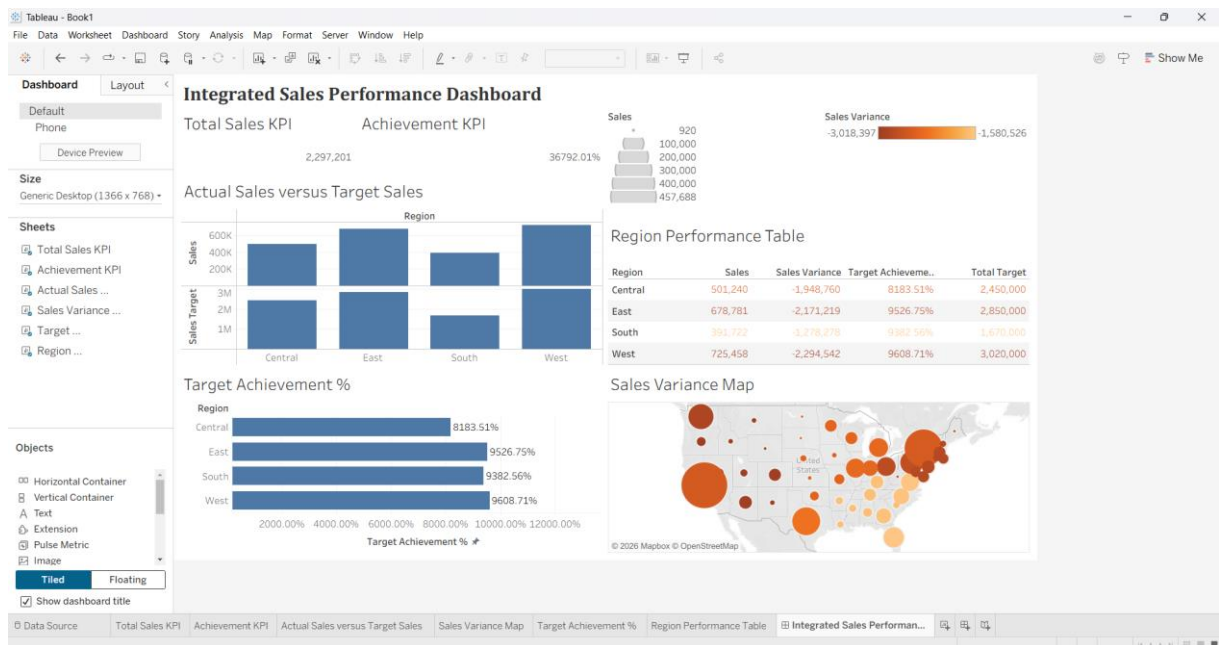
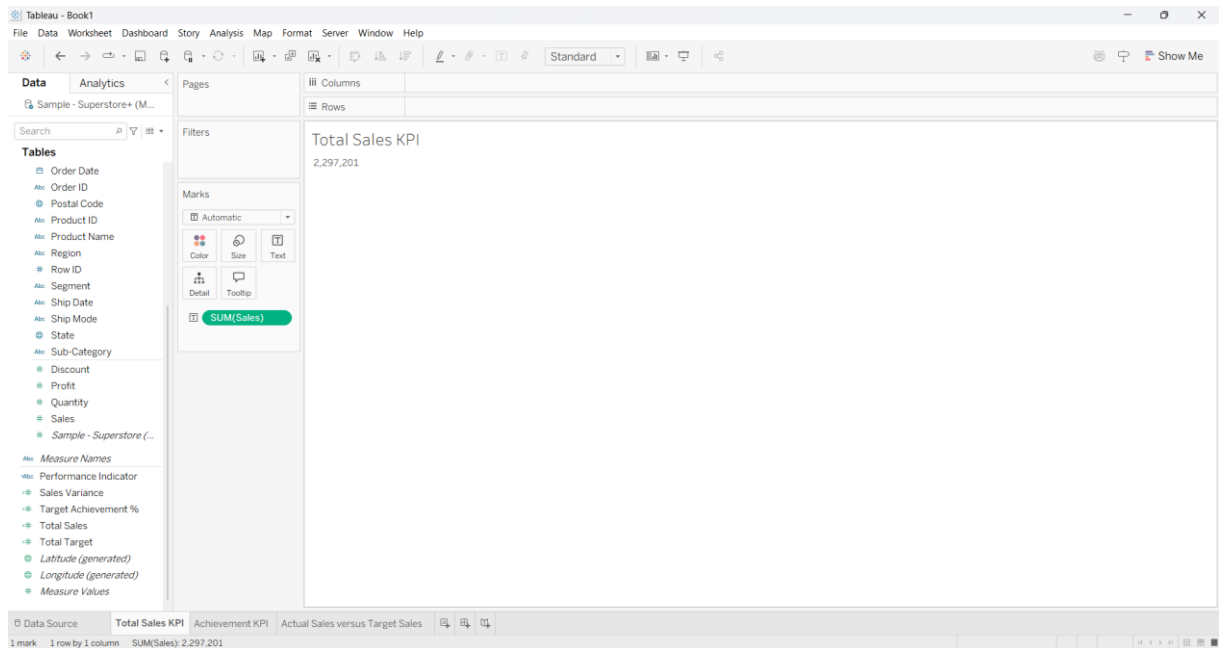
Sample - Superstore

Regional Sales Targets

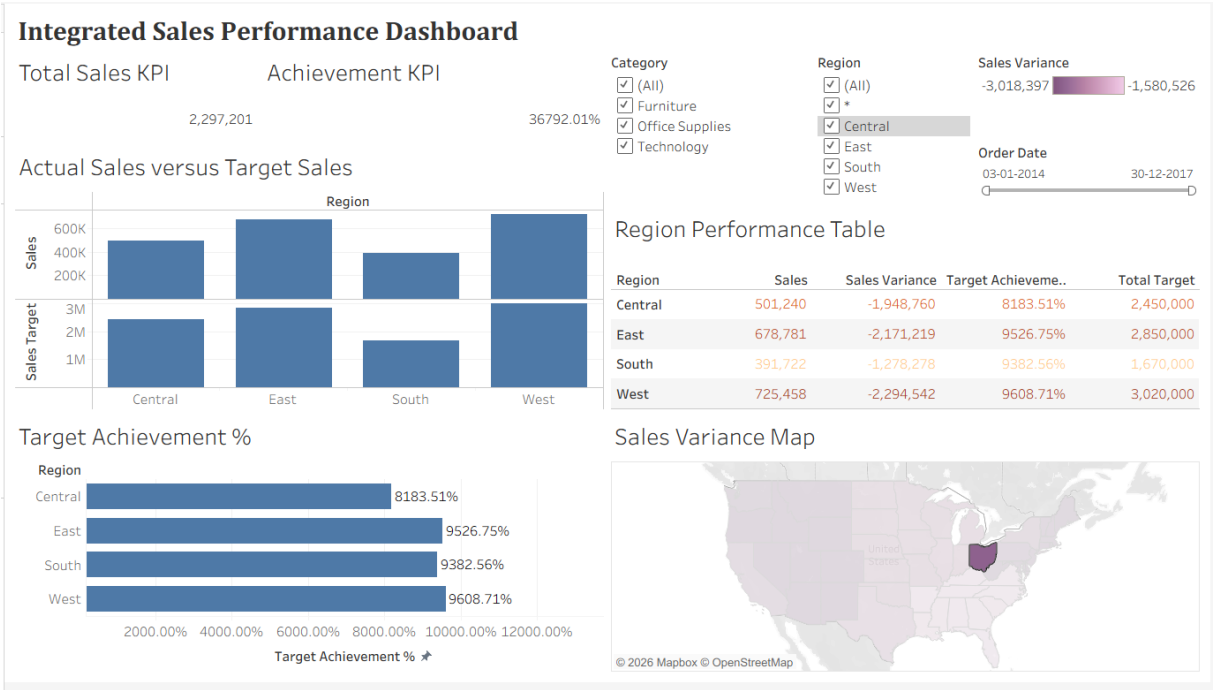
3 fields 16 rows

16 rows

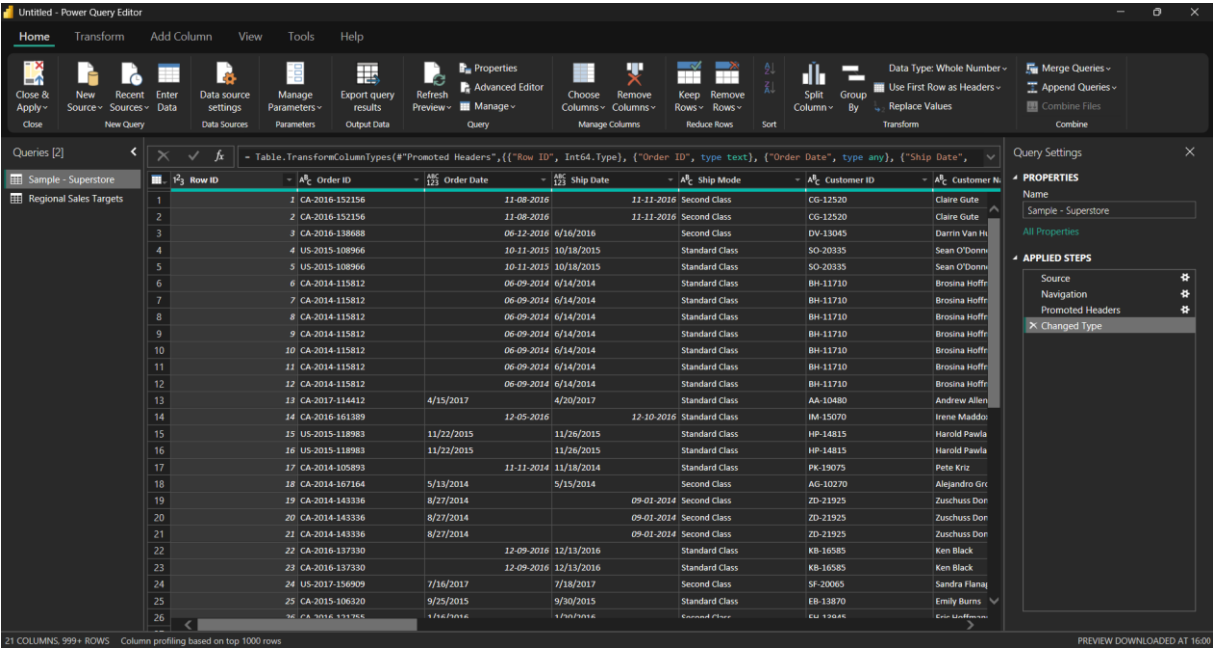
#	Regional Sales Targets	Regional Sales Targets	Regional Sales Targets
Region (Regional Sales Tar	Sales Target	Target Year	
Central	600,000	2026	
East	750,000	2026	
South	450,000	2026	
West	800,000	2026	
Central	650,000	2025	
East	720,000	2025	
South	430,000	2025	
West	770,000	2025	
Central	580,000	2024	
East	700,000	2024	
South	400,000	2024	
West	740,000	2024	
Central	620,000	2023	
East	680,000	2023	

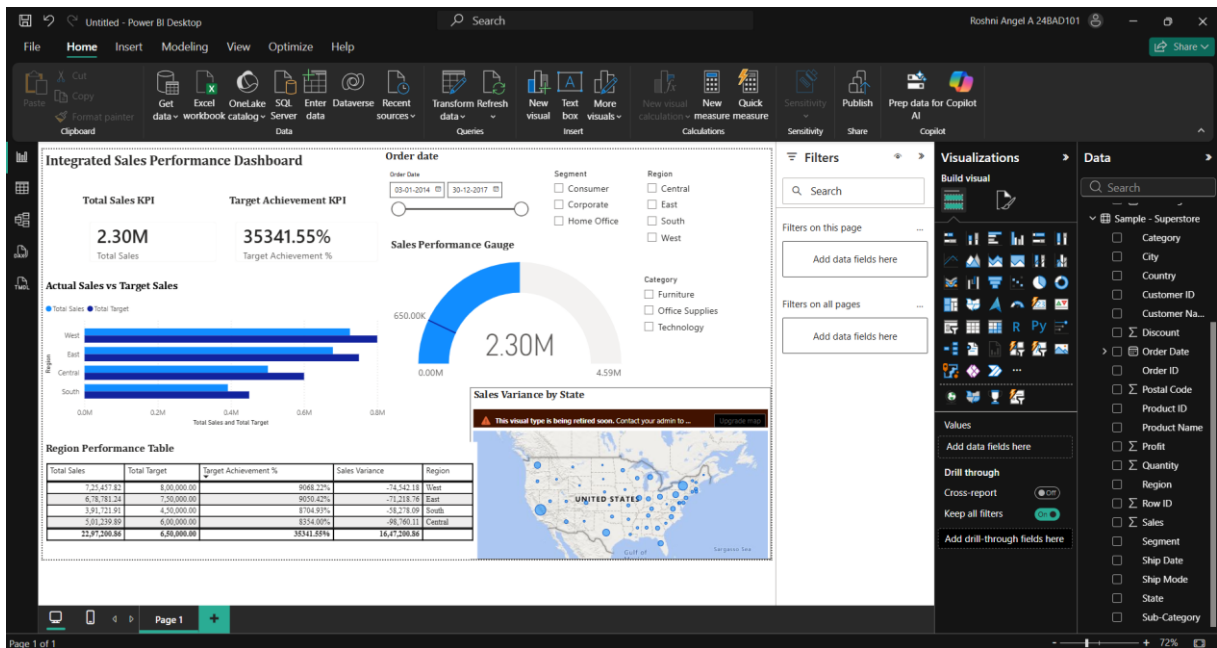
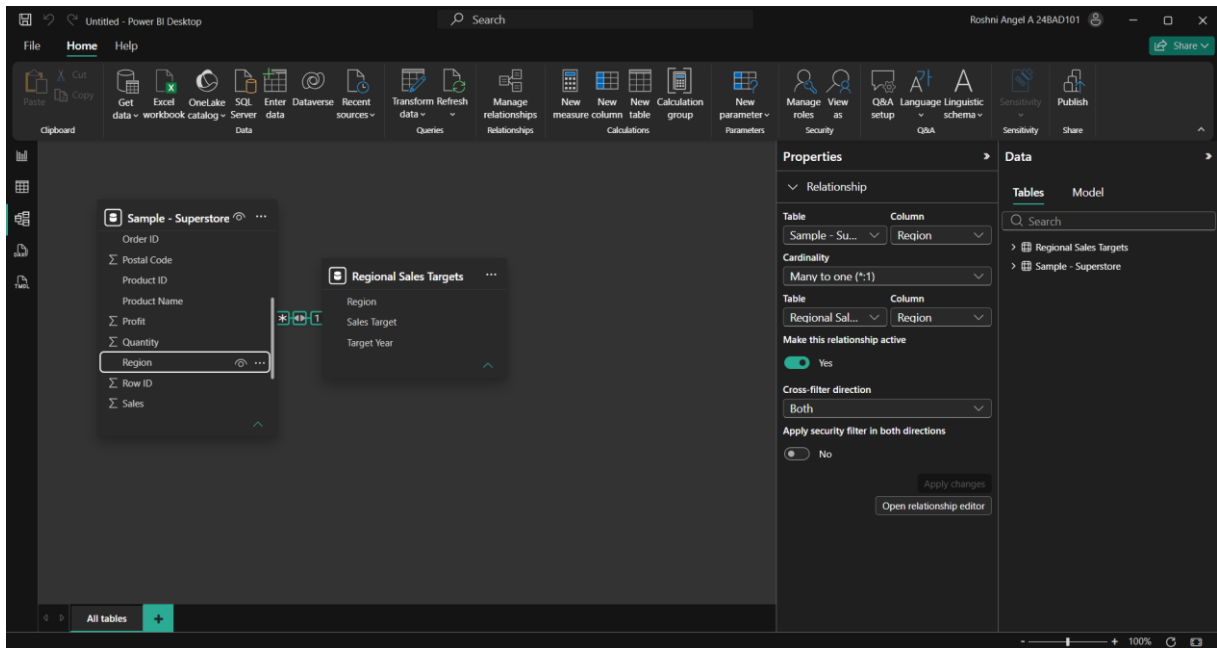


FINAL TABLEAU DASHBOARD :

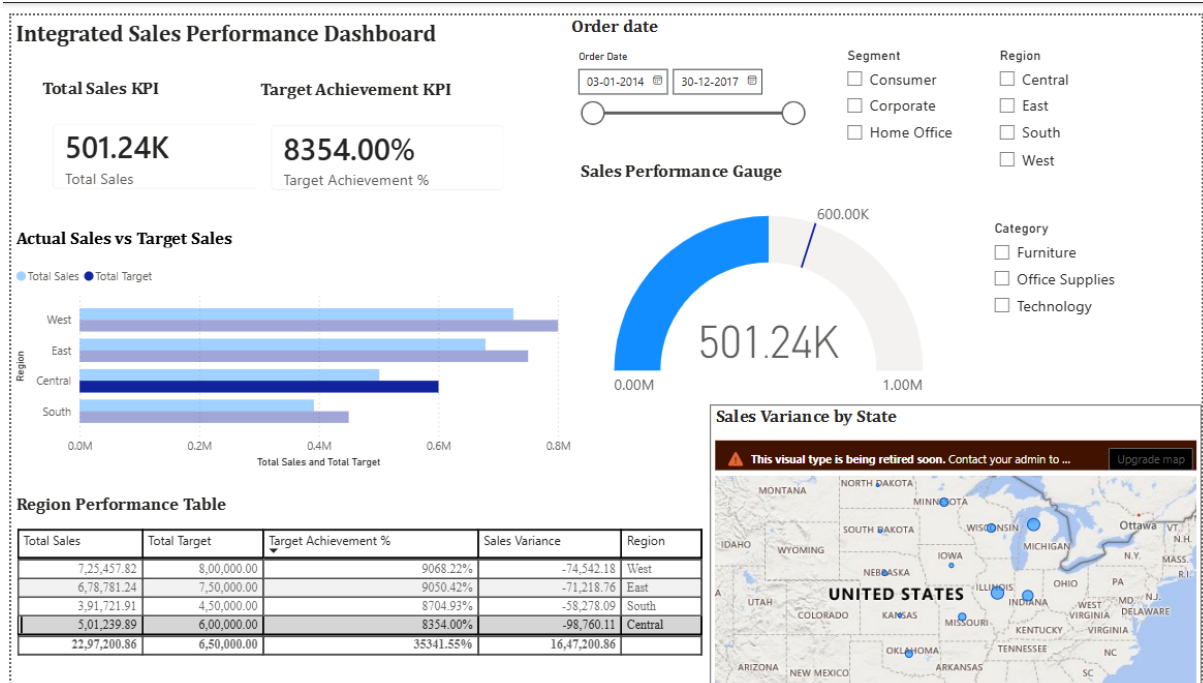


Power BI steps :





FINAL POWER BI DASHBOARD :



POWER BI LINK :

https://kumaragurudtsteam-my.sharepoint.com/:u:/r/personal/roshniangel_24ad_kct_ac_in/Documents/Business%20intelligence/Integrated%20Sales%20Performance%20Dashboard.pbix?csf=1&web=1&e=czdi4Y

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did integrating the sales targets data provide a more complete analytical picture than using the sales data alone?

Integrating the sales targets data with the sales dataset provided a more comprehensive view of business performance. The sales dataset alone showed the revenue generated in each region, but it did not indicate whether the performance met the expected goals. By combining both datasets, it became possible to compare actual sales with target values, calculate variances, and determine the target achievement percentage. This helped identify regions that were performing well and regions that required additional attention.

2. Describe the type of join or relationship you used to combine the two datasets and explain why it was the appropriate choice.

In Tableau, a relationship was created between the two datasets using the **Region** field. In Power BI, a **many-to-one relationship** was established between the **Sample Superstore** table and the **Regional Sales Targets** table. This approach was suitable because a single target value was assigned to each region, while multiple sales records belonged to the same region. Using this relationship allowed information from both datasets to be combined accurately without creating duplicate values.

3.What challenges, if any, did you face regarding data granularity or mismatched keys (e.g., different region names) between the two sources?

One of the main challenges was ensuring consistency between the region names in both datasets. Any differences in spelling, capitalization, or formatting could have produced incorrect results during the integration process. Another challenge was the presence of multiple target values for different years in the target dataset. This issue was resolved by filtering the data and selecting only the records related to the required year. Data types were also verified to ensure that dates, text fields, and numerical values were stored in the correct format.

4.How can creating variance calculations (e.g., Sales vs. Target) help a business manager make better decisions?

Variance calculations help managers understand the difference between actual performance and expected performance. Positive values indicate that the target has been achieved, while negative values highlight areas that require improvement. These calculations enable managers to allocate resources effectively, improve sales strategies, identify underperforming regions, and set more realistic goals for future business operations. As a result, decision-making becomes more efficient and data-driven.

5.What other data source (e.g., marketing campaign data, customer demographics) could you integrate to further enrich this analysis, and what new insights might it provide?

Marketing campaign data could be integrated into the existing analysis to gain a deeper understanding of customer behavior and sales performance. Information such as advertising expenditure, customer responses, and campaign duration could help determine the effectiveness of different marketing strategies. Customer demographic information, such as age, location, and purchasing preferences, could also be included. This additional information would help identify customer segments, predict future trends, and improve overall business performance.

RESULT:

The retail sales and regional targets datasets were successfully connected, cleaned, and integrated in both Tableau and Power BI. By creating relationships between the sources, a unified data model was built. The resulting comprehensive dashboard visualized sales performance against set targets, enabling variance analysis and providing deeper, context-rich insights into regional performance that were not possible with a single data source.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		